



T2R
THEORY TO REALITY

STUDENT RESOURCE · CIVIL ENGINEERING

Introduction to HEC–RAS: Flood Modelling for Civil Engineers

Never heard of flood modelling? No problem. This guide starts from scratch and takes you all the way to understanding how real engineers predict floods — and how you can learn to do it too.

BRANCH

Civil Engineering

LEVEL

Zero to Advanced

PRIOR KNOWLEDGE

None Required

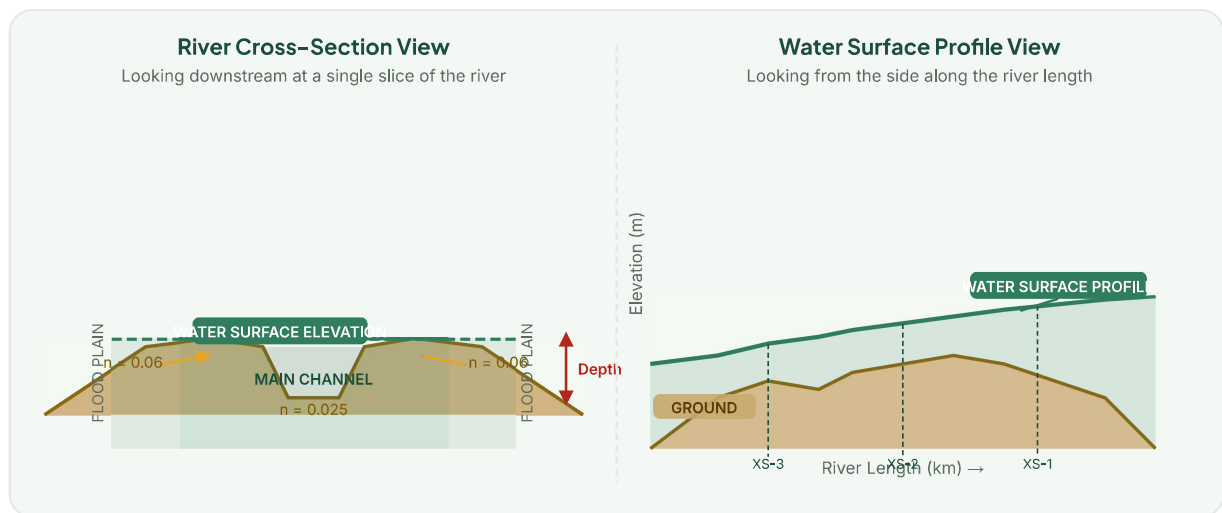
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T2R — Theory to Reality

VISUAL REFERENCE

What HEC–RAS Models Look Like

The diagram below shows the two core outputs of a HEC-RAS model — a **river cross-section** (left) showing how water fills the channel and spills onto the flood plain, and a **water surface profile** (right) showing how the water level varies along the length of the river.



Left: A cross-section showing the river channel, flood plain, water surface elevation, and Manning's n zones. Right: The water surface profile along the river length, with cross-section locations (XS-1, XS-2, XS-3) marked.

LET'S START WITH A REAL SITUATION

Why Does Flood Modelling Even Exist?

Picture this: It is July 2005. Mumbai receives **944mm of rainfall in a single day** — the highest ever recorded in the city's history. Roads disappear. Trains stop. Hundreds of people lose their lives. The city is paralysed for days.

Now here is the question an engineer must answer *before* something like this happens: **If it rains this heavily, where exactly will the water go? How deep will it get? Which areas will flood first?**

You cannot wait for a real flood to find out. It is too dangerous and too late. So engineers use computer programs to **simulate** — or virtually recreate — what a flood would look like, before it ever happens. This allows cities and governments to plan evacuation routes, build drainage systems, design bridges at the right height, and protect lives.

This is what flood modelling is. And HEC-RAS is one of the most important tools engineers use to do it.

💡 Think of it this way

Before a film is released, the director makes a storyboard — a preview of what every scene will look like. Flood modelling is like a storyboard for a flood. The engineer

"films" the flood on a computer before it happens in real life, so everyone can prepare.

THE BASICS FIRST BASIC

How Does Water Actually Flow in a River?

Before we talk about any software, we need to understand what we are trying to model. Let us think about a river.

Water in a river always flows from a **higher place to a lower place** — from the mountains down to the sea. Simple enough. But the speed and depth of that water changes depending on many things:

- **The shape of the river channel** — a wide, flat river flows slowly; a narrow, steep river flows fast
- **The roughness of the riverbed** — a smooth concrete channel lets water flow faster than a rocky, vegetated river bank
- **How much water is coming in** — after heavy rain, much more water enters the river than on a dry day
- **Obstacles in the water** — bridges, dams, weirs, and bends all change how water flows

When there is too much water — more than the river channel can carry — the water spills over the banks and spreads across the surrounding land. **This is a flood.**

✦ Key Vocabulary: Discharge

Engineers measure how much water is flowing in a river using a unit called **discharge**, measured in cubic metres per second (m^3/s). If a river has a discharge of $100 \text{ m}^3/\text{s}$, it means 100 cubic metres of water — roughly 100,000 litres — is passing a fixed point every single second. Major Indian rivers during monsoon can reach tens of thousands of m^3/s .

What Happens During a Flood?

Imagine pouring water into a glass slowly. The water level rises gently. Now imagine pouring the entire bottle at once — the water rises quickly and may overflow the glass. A river behaves exactly the same way during heavy rain. The river level (called

the **water surface elevation**) rises. If it rises above the river bank, water spills out and floods the land around it.

The engineer's job is to predict: **At what level will the water reach? How far will it spread? And how deep will it be?**

INTRODUCING THE TOOL BASIC

So, What Exactly is HEC-RAS?

HEC-RAS stands for **Hydrologic Engineering Center – River Analysis System**. That is a long name — let us break it down:

- **Hydrologic Engineering Center (HEC)** — the organisation that built this software. It is part of the US Army Corps of Engineers, which manages rivers and flood protection across the United States.
- **River Analysis System (RAS)** — a computer program that analyses how water flows through rivers.

HEC-RAS is a **free software** — anyone can download and use it. This is one reason it has become the most widely used river modelling tool in the world, including across India.

In simple terms: **you give HEC-RAS information about a river — its shape, the roughness of its banks, and how much water is flowing — and it tells you what the water level will be at every point along the river.**

Think of it this way

Imagine you have a garden hose and a sandpit. You want to know: if I put this much water into the sandpit, where will it flow and how deep will it get? HEC-RAS is like a computer simulation of that sandpit — but for a real river with real terrain data.



Who Made It?

US Army Corps of Engineers. First released in 1995. Now on version 6.x with regular updates.



How Much Does It Cost?

Completely free to download and use. No license fees. This is why it is so popular worldwide.



Who Uses It in India?

CWC, NHPC, WAPCOS, AECOM, Fichtner, L&T, State Irrigation Departments, and hundreds of consulting firms.



Do Students Learn It?

Rarely in college — which is exactly why knowing it gives you a big advantage in placements and internships.

GOING DEEPER INTERMEDIATE

What Types of Analysis Can HEC-RAS Do?

HEC-RAS can simulate rivers in different ways depending on what the engineer needs to know. Think of it like different camera angles for the same scene.

1D vs 2D Analysis — What's the Difference?

This sounds technical, but the concept is simple:

- **1D (One-Dimensional) Analysis** — The software looks at the river from above, like a long line. It calculates the water level at specific "slices" of the river, called cross-sections. Think of it like measuring the water depth at specific points along a road — you know the depth at each point, but not exactly what happens in between. 1D is faster to set up and works well for long stretches of rural river.
- **2D (Two-Dimensional) Analysis** — The software treats the entire flood plain as a surface, like a map, and simulates water spreading across it in all directions. This is more detailed and more accurate for cities, deltas, and areas where water can flow in many directions at once.

💡 Think of it this way

1D analysis is like measuring temperature at a few cities along a highway and drawing a graph. 2D analysis is like having a full weather satellite image showing the temperature across the entire country. Both are useful — it depends on how much detail you need.

Steady Flow vs Unsteady Flow

This is another important distinction:

- **Steady Flow** — The amount of water in the river stays the same throughout the analysis. The engineer picks one flood size (e.g. the worst flood expected in 100 years) and asks: "What does the river look like during this flood?" This is used for designing bridges and checking whether embankments are tall enough.
- **Unsteady Flow** — The amount of water changes over time, just like in a real flood. The model tracks how the flood wave builds up, moves downstream, and then recedes. This is more realistic and is used for dam break studies, flood warning systems, and detailed flood routing.

CORE CONCEPTS INTERMEDIATE

The Building Blocks of a HEC-RAS Model

Every HEC-RAS model is built from a few key pieces of information. Let us go through each one simply.

1. Cross-Sections — The Shape of the River

Imagine you could freeze the river and slice through it like a loaf of bread. The shape of that slice — how wide it is, how deep, what the banks look like — is called a **cross-section**. HEC-RAS uses dozens or hundreds of these cross-sections, spaced along the river, to understand the shape of the entire channel.

Where does this data come from? Engineers survey the river using instruments like Total Station, GPS, or more recently, drones and LiDAR (a laser-based measurement technology). For larger rivers in India, satellite terrain data like Cartosat DEM or SRTM is also used.

2. Manning's Roughness Coefficient (n) — How Rough is the Ground?

Water flows faster over a smooth surface than a rough one. Think about the difference between water flowing over a smooth tile floor versus flowing through a field of tall grass. Engineers use a number called **Manning's n** to represent this roughness.

Surface Type	Manning's n Value	Water Behaviour
Smooth concrete channel	0.012	Water flows very fast
Clean, straight earthen channel	0.022	Moderate speed
Natural river, some vegetation	0.035	Slower

Choosing the right Manning's n value is one of the most important skills in flood modelling — and it comes with experience. Getting it wrong can make your flood predictions dangerously inaccurate.

3. Boundary Conditions — What's Happening at the Edges?

HEC-RAS needs to know what the water is doing at the very start and very end of your model. These are called **boundary conditions**.

- **Upstream boundary** — How much water is entering the river? (discharge in m^3/s)
- **Downstream boundary** — What is the water level at the end of the model? (Often the sea level, a reservoir, or a known water level)

4. Water Surface Profile — The Output

After you run the model, HEC-RAS produces a **water surface profile** — a line showing the predicted water level at every point along the river. The engineer then compares this to the surrounding ground levels to see which areas will flood, and how deep the water will be. This is the answer to the original question: "Where will the water go?"

THE FULL PROCESS INTERMEDIATE

How is a Real HEC–RAS Model Built? Step by Step.

1

Collect Terrain Data

The first step is getting accurate information about the shape of the land — the river channel and the surrounding flood plain. In India, engineers use survey instruments for detailed local studies, and freely available satellite data like **SRTM (30m resolution)** or **Cartosat DEM** for regional studies. For urban areas, drone surveys provide very high accuracy.

2

Define the River Geometry in HEC–RAS

Using the terrain data, the engineer draws the river centre line and extracts cross-sections in HEC-RAS. Any structures that affect flow — bridges, dams, weirs, embankments — are also added at this stage. A bridge across the river, for example, narrows the channel and forces water to flow faster underneath it. HEC-RAS accounts for this automatically.

3

Determine the Design Flood Discharge

How much water should the model simulate? Engineers use historical rainfall and river gauge data (collected by the **Central Water Commission** across India) to estimate the flood that could happen once in every 25, 50, 100, or 500 years. This is called the **return period flood**. A bridge is typically designed to survive a 100-year flood without being damaged.

4

Set Boundary Conditions and Run the Model

Enter the design flood discharge at the upstream end and the known water level at the downstream end. Then click Run. HEC-RAS solves the hydraulic equations for every cross-section and produces the water surface profile in seconds.

5

Calibrate the Model

Before trusting the model's predictions, the engineer checks them against **real past flood data**. If a major flood occurred in 1998, the engineer runs the model for that flood and checks if the predicted water levels match the actual recorded levels. If they don't match, the Manning's n values and geometry are adjusted until the model is accurate. This is called **calibration** — and it is what makes a model trustworthy.

6

Map the Flood Zones

Finally, the engineer exports the water surface elevations to a GIS software (like **QGIS** or **ArcGIS**) and overlays them on the terrain map. The result is a **flood inundation map** — a colour-coded map that shows exactly which areas will be underwater, and how deep the water will be, during a given flood event. This map is what city planners, disaster management officials, and insurance companies use to make decisions.

IN Real Indian Example: Brahmaputra River, Assam

The Brahmaputra is one of the world's largest rivers and causes catastrophic floods in Assam almost every year. Engineers at CWC and WAPCOS use HEC-RAS to model flood levels along the Brahmaputra, helping the Assam State Disaster Management Authority plan evacuations before the monsoon even peaks. The 2D models they build now cover hundreds of kilometres of the river and its flood plain, updated each year with new terrain data.

GOING FURTHER **ADVANCED**

Advanced Capabilities of HEC-RAS

Once you have mastered the basics, HEC-RAS can do much more sophisticated analysis. Here is a glimpse of what experienced engineers use it for:

Dam Break Analysis

What happens if a dam fails suddenly? How fast does the flood wave travel? Which villages downstream are at risk, and how many minutes do they have to evacuate? HEC-RAS can simulate this scenario using unsteady flow analysis, and the results are used to create **Emergency Action Plans (EAPs)** for every large dam in India.

Sediment Transport Modelling

Rivers do not just carry water — they carry sediment (sand, silt, gravel). Over time, sediment deposits in reservoirs (reducing their capacity) and gets eroded from river banks (destroying infrastructure). HEC-RAS has a dedicated sediment module that models this over years and decades — critical for the long-term management of Indian reservoirs like Bhakra Nangal and Hirakud.

Water Quality Analysis

HEC-RAS can also model how pollutants — such as industrial discharge or sewage — spread through a river system. This is increasingly important in India's efforts to clean rivers like the Ganga (Namami Gange Programme) and the Yamuna.

Real-Time Flood Forecasting

In modern flood management systems, HEC-RAS models are connected to live rainfall data and river gauge readings. The model runs automatically every few hours and sends flood predictions to authorities — giving communities advance warning of incoming floods. This is now operational in parts of Odisha, Bihar, and West Bengal.

YOUR LEARNING PATH

How to Start Learning HEC-RAS — Practically, Right Now

You do not need any special equipment or paid courses. Here is exactly what to do:

1 Download HEC-RAS (Free)

Go to hec.usace.army.mil and download HEC-RAS version 6.x. It works on Windows. The download includes HEC-RAS Mapper, which lets you visualise your flood maps on a satellite image.

2 Download QGIS (Free)

Go to **qgis.org** and download QGIS. This is a free GIS software that lets you work with maps, terrain data, and flood maps. HEC-RAS + QGIS is the standard combination used at most firms in India.

3 Follow the Official Tutorial — Bald Eagle Creek

HEC-RAS comes with built-in example projects. Start with "Bald Eagle Creek" — it teaches you a complete 1D steady flow model from scratch in about 2–3 hours. Every step is guided. This is how most professionals also learned.

4 Download Real Terrain Data for a River Near You

Go to **earthexplorer.usgs.gov** (free account required). Search for your city or a nearby river. Download the SRTM 1 Arc-Second DEM data. Import it into QGIS. You now have real terrain data for your own flood model — no need to make anything up.

5 Build Your First Real Project

Pick a small river or drainage channel near your college. Use the SRTM data to extract cross-sections. Estimate a design flood discharge using data from the CWC website. Run the model and produce a simple flood inundation map in QGIS. This small project, done independently, is exactly the kind of thing that impresses recruiters at placement interviews.

How to Present This on Your Resume

- Developed a 1D HEC-RAS steady flow model for [River Name] river, [City], using SRTM terrain data
- Performed flood frequency analysis to estimate 25-year and 100-year return period discharges
- Produced flood inundation maps in QGIS identifying at-risk areas and estimated flood depths
- Calibrated Manning's roughness coefficients against available historical water level data

REAL WORLD

Where HEC–RAS is Used in India

Project Type	Application of HEC-RAS	Where in India
Bridge Design	Calculating minimum bridge deck height above flood level	National Highways everywhere
Flood Zone Mapping	Identifying areas at risk for urban planning	Ganga, Brahmaputra, Krishna basins
Dam Safety Studies	Dam break analysis for emergency action plans	Uttarakhand, J&K, Himachal
Irrigation Infrastructure	Canal capacity design and overflow structure sizing	AP, Maharashtra, Punjab
Urban Flood Management	Stormwater drain sizing and flood risk mapping	Chennai, Hyderabad, Mumbai
Hydropower Projects	Backwater analysis upstream of dam	Himalayas, Western Ghats

CAREER PATHS

What Jobs Can You Get With This Skill?

Hydraulic modelling is a niche skill — most Civil Engineering graduates in India do not have it. That makes you stand out immediately. Roles that use HEC-RAS directly include:

- Hydraulic Engineer
- Flood Risk Analyst
- Water Resources Engineer
- Dam Safety Engineer
- Urban Drainage Engineer
- Disaster Management Consultant
- GIS & Remote Sensing Analyst
- Environmental Impact Assessor

Key employers in India: WAPCOS, AECOM India, Fichtner Water & Transportation GmbH, NHPC, CWC, State Irrigation Departments, RITES, L&T Infrastructure, Jacobs Engineering, Stantec, Halcrow (now CH2M).

SUMMARY

Key Takeaways

- **Flood modelling exists** because engineers need to predict where water will go before a real flood happens — to design infrastructure, plan evacuations, and protect communities
- **HEC-RAS** is the world's most widely used flood modelling tool, completely free, and heavily used across India's water sector
- The core inputs are: **river geometry** (cross-sections), **Manning's n** (roughness), **design flood discharge**, and **boundary conditions**
- The core output is a **water surface profile** — which becomes a flood inundation map when overlaid on terrain in GIS software
- **Calibration** is what makes a model trustworthy — always validate against real historical flood data
- You can start learning today with **zero budget** — HEC-RAS and QGIS are both free, and real terrain data is freely available online
- Building even one small self-taught project on a local river creates a **significant advantage** at placement interviews